

OM BHATT

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PROFESSIONAL SUMMARY

Well-rounded student with strong academics and extracurricular experience. Passionate about robotics, particularly autonomous and intelligent systems. Seeking to expand engineering knowledge and experience to design, develop, and optimize mechatronic and robotic systems that advance automation and contribute to the betterment of society.

U.S. Person (No Sponsorship Required)

EDUCATION

Texas A&M University, *B.S - Mechatronics Engineering*
Minor: Embedded Systems

2024 - 2028
College Station, TX

Relevant Coursework: Calculus I–III, Differential Equations, Electricity & Magnetism, Engineering Mechanics, Strength of Materials, Metallic Materials, Fluid Mechanics & Power, Circuit Analysis, Analog Electronics, Digital Electronics, Embedded Systems Development in C, Microcontroller Architecture, Applied Dynamic Systems

Certifications: Introduction to Programming using Python

Organizations: IEEE TAMU, ASME TAMU, TURTLE Robotics, Texas Aimbots, SOMTECH, TAMUHACK, SASE

RELEVANT EXPERIENCE

R&D Robotics Engineer - Software Subteam

Sep 2025 - Present

T.U.R.T.L.E Robotics, Disaster Response Observation Network, Texas A&M University

College Station, TX

- Researched and programmed autonomous flight algorithms (offboard_control_square_2) using PX4 and MAVROS over ROS 2 Humble to command UAVs for a 10-member team's autonomous UAV swarm platform assisting first responders during structural fires.
- Co-authored and presented the "[Disaster Response Observation Network \(DRON\)](#)" research poster across two university showcases (Nov 2025, Apr 2026), detailing autonomous swarm progress and system architecture to an audience of engineering faculty and peers.

Hardware Team Member

Sep 2025 - Present

Texas Aimbots, Texas A&M University

College Station, TX

- The [Sentry Robot](#) is a fully autonomous ground robot that moves and shoots at enemy robots using computer vision.
- Designed 4+ structural components, including the gimbal housing, in Onshape CAD for Sentry, a fully autonomous ground robot capable of traversing the field, engaging targets, self-healing, and operating without driver input.
- Integrated the NVIDIA Jetson Orin, LiDAR, and camera systems to enable real-time perception and autonomous decision-making ahead of the 2026 ARC Championship, an international collegiate robotics competition.

AEOP Summer Robotics Research Intern - LLMs for Safe Navigation REU

June 2025 - Aug 2025

Oden Institute - Center for Autonomy - Autonomous Systems Group, UT Austin

Austin, TX

- Researched autonomous off-road navigation and drift/jump control dynamics under [Dr. Ufuk Topcu](#), [Dr. Christian Ellis](#), and [PhD Student Rwik Rana](#) to architect a low-cost, 1/16 scale autonomous RC car as a modular research platform for drift and jump navigation studies.
- Executed the complete electromechanical assembly by designing and soldering custom circuit boards to successfully power and integrate all components, including a Raspberry Pi 5/Jetson Nano and motor controllers.

- Programmed a robust low-level controller in a C++/ROS 2 software stack to achieve precise Ackerman steering and throttle control for the vehicle.
- Established full teleoperation capabilities and utilized the F1Tenth simulator to validate SLAM and control algorithms with IMU fusion for post-jump re-localization and improved drift control, preparing the platform for fully autonomous operation.

Hatchling Team Member

T.U.R.T.L.E Robotics, Texas A&M University

Jan 2025 - Sep 2025

College Station, TX

- Won 2nd Place out of 30+ competing teams at the [2025 TURTLE Robotics Hatchling Competition](#)
- Designed and CADded key mechanical components in SolidWorks, including a forklift-style lift and gripper mechanism engineered for rapid block acquisition.
- Programmed an ESP32 and Arduino to control precise vehicle navigation and operate the multi-stage lift, enabling the robot to successfully retrieve and stack multiple blocks.
- Collaborating with the T.U.R.T.L.E Robotics team at Texas A&M University to enhance robotics expertise and tackle innovative challenges.

Student Researcher: Engineering Science Fairs - Dual-axis Solar Tracker

Science Fair Club, Vista Ridge High School

Sep 2022 - Mar 2024

Cedar Park, TX

- Engineered and developed "[EcoTrack](#)", a novel phototropic solar tracker inspired by nature, which uses thermo-responsive polymers to follow a light source. This design eliminates the need for electrical components in the tracking mechanism, creating a low-cost and low-maintenance solution for developing nations.
- Validated the system through experimentation, demonstrating that EcoTrack harnessed 87 percent more net energy than a traditional dual-axis tracker and 119 percent more than a fixed panel system under identical conditions.
- Achieved [first place](#) across Austin region at Austin Regional Science Fair and advanced to TXSEF State Fair
- Awarded a university tuition scholarship from [Jacobs Engineering](#) (\$2000) and [RICOH Sustainable Development Award](#)
- Elected to lead the role of an officer to help mentor other students on their projects and showcase our school's achievements to attract potential sponsors

Mechatronics Research Intern - WATonoBus: Autonomous Shuttle

Mechatronics Vehicle Systems Lab, University of Waterloo

Jun 2021 - Sep 2021

Waterloo, Canada

- Worked under the guidance of [Professor Amir Khajepour](#) on [MVS Lab](#)'s autonomous shuttle called [WATonoBus](#)
- Designed, fabricated, and programmed an [LED display module](#) that helped inform non-ego agents about the shuttle's intentions
- Awarded a scholarship from MVS Lab for demonstrated excellence
- Completed over 60+ hours of development

PUBLICATIONS

- I. Wilhite, C. Ambroziak, A. Briggs, R. Kato, **O. Bhatt**, L. Breedlove, V. So, R. Shah, M. Shi, J. Shouba, O. Thomas, C. Weatherspoon, and D. Wheaton, "Disaster Response Observation Network (DRON)," TURTLE Robotics, Apr. 2026. [Conference Poster].
- I. Wilhite, A. Briggs, C. Ambroziak, R. Kato, D. Wheaton, **O. Bhatt**, A. Talal, J. Shouba, C. Xu, O. Thomas, and S. Ahn, "Disaster Response Observation Network (DRON)," TURTLE Robotics, Nov. 2025. [Conference Poster].

- **O. Bhatt**, "Autonomous RC Rally Car for Jump and Drift Racing," under the guidance of Dr. U. Topcu, Dr. C. Ellis, and R. Rana, AEOP Summer Robotics Research REU, UT Austin, Aug. 2025. [Research Presentation & Abstract]

VOLUNTEERING AND LEADERSHIP

Autodesk Ambassador Program

Autodesk, Texas A&M University

Jan 2026 - Present

College Station, TX

- Selected for a competitive ambassador program to represent Autodesk, promoting CAD tools through campus events and workshops.

Design Review Officer

T.U.R.T.L.E Robotics, Project Branch, Texas A&M University

Jan 2026 - Present

College Station, TX

- Evaluate system requirements, CAD/electronics documentation, and BOMs across 20+ projects over 3 Design Review sessions.

Sponsorship Committee Member

T.U.R.T.L.E Robotics, External Branch, Texas A&M University

Sept 2025 - Jan 2026

College Station, TX

- Securing corporate sponsorships for TURTLE, a student-led organization, through a new Corporate Sponsor Outreach Initiative.
- Developing and executing outreach strategies to engage with companies and secure funding for workshops, social events, and advanced technical projects.

Texas Science and Engineering Fair Judge

College of Engineering, Texas A&M University

Mar 2025 - Present

College Station, TX

- Evaluate 20+ student research projects annually in the Energy: Sustainable Materials & Designs (2025) and Robotics & Intelligent Machines (2026) categories at the Texas Science and Engineering Fair [TXSEF](#), providing constructive feedback to the next generation of engineers.
- As part of Judge appreciation, George H.W. Bush Combat Development Complex gave TXSEF 2025 judges a tour of the Research Integration Center, which has state-of-the-art technology.

Student Council Member

Student Council Club, Vista Ridge High School

Sep 2022 - Sep 2024

Cedar Park, TX

- Contributed over 20 hours of community service by participating and helping organize events including school Christmas decorations, district events, conducting fundraisers, elementary school outreach events and more
- Donated items such as toys and food for charities

SKILLS

Robotics & Autonomy: ROS 2, PX4, MAVROS, SLAM, OpenCV, PWM/PID Control, Teleoperation, Gazebo, RViz, Point Clouds

Hardware and Prototyping: ESP32, Arduino, Raspberry Pi, STM32, FPGA, LiDAR, Cameras, PCB Design, Soldering, Analog Electronics, Oscilloscope & DMM Testing, Data Acquisition Systems

Programming Languages: Python, C/C++, Embedded C, Linux, Shell (Bash/Zsh), $\text{\LaTeX}$, HTML, CSS

Simulation and CAD: SolidWorks, FEA, Autodesk Inventor, Onshape, Bambu Studio, Eagle, Multisim, MATLAB/Simulink

AI & Software Tools: Git/GitHub, Visual Studio Code, Workspace, Microsoft Office, LLM Integration (ChatGPT, Claude, Gemini)

AWARDS

AEOP Summer Robotics Research REU Scholarship (\$6000) , Oden Institute	Jul 2025
Presented by Army Educational Outreach Program (AEOP) and Oden Institute at the University of Texas at Austin for my summer REU work.	
2nd Place - 2025 TURTLE Robotics Hatchling Competition , T.U.R.T.L.E Robotics	May 2025
Won 2nd Place at the 2025 TURTLE Robotics Hatchling Competition.	
AP Scholar , College Board	May 2024
Granted to students who receive scores of 3 or higher on three or more AP Exams.	
Vista Ridge High School Scholar: “A” Honor Roll , Vista Ridge High School	May 2024
Presented to students who have straight As at the end of the school year in all classes enrolled	
Texas Science & Engineering Fair Finalist , TXSEF	Mar 2024
Advanced to the Texas Science & Engineering Fair (Statewide Science Fair)	
2nd Place at Austin Regional Science Fair , Austin Energy	Feb 2024
Placed second in the Energy: Sustainable Materials and Design category across the Austin metropolitan region	
Vista Ridge High School Scholar: “A” Honor Roll , Vista Ridge High School	May 2023
Presented to students who have straight As at the end of the school year in all classes enrolled	
Texas Science & Engineering Fair Finalist , TXSEF	Mar 2023
Advanced to the Texas Science & Engineering Fair (Statewide Science Fair)	
1st Place at Austin Regional Science Fair , Austin Energy	Feb 2023
Placed first in the Energy: Sustainable Materials and Design category across the Austin metropolitan region	
Jacobs Engineering Tuition Scholarship (\$2000) , Jacobs Engineering	Feb 2023
Presented to one selected finalist as a tuition scholarship that can be applied to the college of choice	
RICOH Sustainable Development Award , RICOH	Feb 2023
Presented to the student whose research has demonstrated the principles and technical innovations that offer the greatest potential for sustainable development	
Summer Intern Scholarship , University of Waterloo	Jun 2021
Presented by Professor Amir Khajepour to me for my work at Mechatronics Vehicle Systems Lab	